

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Designed and shipped to production a self-service analytics agent over 67 BigQuery tables with sub-agent orchestration and context-isolated validation (Context Researcher → SQL Drafter → Validator → Devil's Advocate); 8 deterministic SQL rules and a 23-case golden-query evaluation suite as guardrails. Owned it concept to launch. Approved by Sr. Director and product leadership and currently the only AI skill in active use by business teams on Data Ventures.
- Built an autonomous Jira ticket-resolution agent: 6-gate triage, MCP-driven context gathering, FAISS+BM25 retrieval over 11,000 historical SQL queries, self-healing execution with a 10-retry budget. Closed 400+ data requests across 8 categories over 15 months and cut turnaround from 30–60 minutes per request to under 10 minutes each. Stakeholders never knew an agent was answering them.
- Built a headless KPI monitor on a 6-hour cron that detects threshold breaches, runs root-cause analysis, and posts Slack alerts; decision logs and state persist to BigQuery for end-to-end observability, so the agent has no LLM-memory reliance. Designed a v2 as a 3-tier event-driven control plane self-configuring via a BigQuery event registry, so new event types ship without code deploys.
- Built a registry-governed semantic layer over BigQuery that boxes the LLM into semantic parsing (one call, strict JSON intent, never SQL) with deterministic validation, value canonicalization, and a deterministic SQL compiler; new or changed definitions pass a human review gate before entering the registry.
- Built a hybrid orchestrator on top of the analytics agent that lets Product self-serve requests they previously filed as tickets; routes questions through a semantic-layer skill and the agent, reasons about disagreements, and feeds discrepancies back to refine definitions.
- Delivered 5 Power BI dashboards ranked top 1.4% of Walmart's 207,000 dashboards (1,064 views, 22 users); cut data-request turnaround from 1–24 hours to 5–10 minutes; Making a Difference Award.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment; controlled A/B tests persuaded 15% to donate to an opposing cause.

FTI Consulting — UChicago MSADS Capstone

Chicago, IL · Jan – Aug 2024

- RAG-to-proposal generator deployed on AWS EC2; cut proposal time ~30%; Best in Show — UChicago MSADS Capstone.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit + Gemini 2.5 + LangChain-orchestrated RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Agent stack: MCP server design, sub-agent orchestration, context engineering, prompt engineering, RAG (FAISS + BM25 + RRF), evaluation suites (golden sets, human-in-the-loop validation), multi-model routing across providers, headless cron agents, event-driven architectures, decision logging / observability

APIs / backend / glue: FastAPI, Cloud Run, BigQuery scheduled queries, Slack webhooks, cron, Playwright SSO, JSONL state stores, CI/CD (pytest, version control)

LLMs: Claude (Opus / Sonnet / Haiku via Wibey), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI

Data: BigQuery, dbt, SQL, Python, PySpark, Airflow, Databricks, MySQL

Viz: Streamlit, React Recharts, Power BI (DAX, XMLA TMSL), Tableau, LookML, Matplotlib

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL